Abstract Algebra

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Contents

1 Group 1
1.1 Group and Subgroup 1
1.2 Coset and Conjugacy 4
1.3 Group Homomorphism 7
1.4 Permutation and Group Action 9
1.5 Sylow Theorem 11
1.6 Finitely Generated Abelian Group 14
2 Ring 16
2.1 Ring Basics 16
2.2 Ideal and Quotient Ring 18
2.3 Unique Factorization 21
3 Field 25
3.1 Field Extension 25
3.2 Splitting Field 26
3.3 Finite Field 28

1 Group

1.1 Group and Subgroup

Definition 1.1. *(Group)*
Let G be a nonempty set with a binary operation (the operation is closed). If

- $(ab)c = a(bc)$ for all $a, b, c \in G$;
- there exists $e \in G$ such that $ea = a = ae$ for all $a \in G$;
- for every $a \in G$, there exists $b \in G$ such that $ba = e = ab$;

then $(G, \cdot)$ is a group. The element e is the identity, and b is the inverse of a, denoted by a^{-1}.

Definition 1.2. *(Semigroup)*
A nonempty set with an associative binary operation is a semigroup.

Definition 1.3. *(Abelian Group)*
A group $(G, \cdot)$ is abelian if $ab = ba$ for all $a, b \in G$.

Definition 1.4. *(Order)*
The order of a group G is the number of its elements, denoted by $|G|$.

Remark.
- $(ab)^{-1} = b^{-1}a^{-1}$.
- The identity of a group is unique, and the inverse of each element is unique.
- After a definition, one should check that it is well-defined.

Definition 1.5. *(Left Identity)*
Let G be a semigroup. An element $e_l \in G$ is a left identity if $e_l g = g$ for every $g \in G$. An element a_l^{-1} is a left inverse of a if $a_l^{-1}a = e_l$.

Proposition 1.1.
A semigroup G is a group if and only if it has a left identity and every element has a left inverse.

Proof.
The forward direction is immediate. Conversely, let e be a left identity, and for each a let a_l^{-1} satisfy $a_l^{-1}a = e$. The left inverse of a_l^{-1} then gives

$$aa_l^{-1} = e(aa_l^{-1}) = \left((a_l^{-1})_l^{-1}a_l^{-1}\right)(aa_l^{-1}) = (a_l^{-1})_l^{-1}(a_l^{-1}a)a_l^{-1} = (a_l^{-1})_l^{-1}a_l^{-1} = e.$$

Thus a_l^{-1} is also a right inverse. Moreover

$$ae = aa_l^{-1}a = ea = a,$$

so e is a right identity, and G is a group. ■

Proposition 1.2.
A finite semigroup G is a group if and only if it satisfies the left and the right cancellation laws.

Proof.
The forward direction is trivial. Conversely, left cancellation makes $L_a : x \mapsto ax$ injective, hence bijective. Choose e_l with $ae_l = a$. For every b,

$$ae_l b = ab,$$

so left cancellation yields $e_l b = b$. Thus e_l is a left identity. Likewise $R_a : x \mapsto xa$ is bijective, so each a has a left inverse relative to e_l. The previous proposition shows that G is a group. ■

Definition 1.6. *(Subgroup)*
Let $(G, \cdot)$ be a group. A nonempty subset $H \subseteq G$ is a subgroup of G if $(H, \cdot)$ is a group, denoted by $H \leq G$.

Remark.
A subgroup has the same identity as the original group.

Proposition 1.3.
Let H be a nonempty subset of a group G. Then $H \leq G$ if and only if $ab^{-1} \in H$ whenever $a, b \in H$.

Proof.
If H is a subgroup, then $a, b \in H$ gives $b^{-1} \in H$ and $ab^{-1} \in H$. Conversely, take $a = b$ to get $e = aa^{-1} \in H$. Take $a = e$ to get $b^{-1} \in H$. Replacing b by b^{-1} then gives $ab \in H$. ■

Remark.
If $H \leq G$, then $H^{-1} = H = HH$.

Proposition 1.4.
Let $A, B \leq G$. Then $AB \leq G$ if and only if $AB = BA$.

Proof.
If AB is a subgroup, then $AB = (AB)^{-1} = B^{-1}A^{-1} = BA$. Conversely, assume $AB = BA$. Since A and B are subgroups,

$$AB(AB)^{-1} = ABB^{-1}A^{-1} = ABBA = ABA = BAA = BA = AB.$$

Thus $AB(AB)^{-1} = AB$, so $AB \leq G$. ■

Definition 1.7. *(Center)*
The center of a group G is

$$C(G) = \{x \in G \mid \forall g \in G, xg = gx\}.$$

Definition 1.8. *(Centralizer)*
For $a \in G$, the centralizer of a is $C_G(a) = \{g \in G \mid ag = ga\}$. For $H \subseteq G$,

$$C_G(H) = \{g \in G \mid \forall h \in H, gh = hg\}.$$

Definition 1.9. *(Normalizer)*
For $H \subseteq G$, the normalizer of H is $N_G(H) = \{g \in G \mid gH = Hg\}$.

Remark.
$C(G)$, $C_G(a)$, $C_G(H)$, and $N_G(H)$ are all subgroups of G.

Definition 1.10. *(Subgroup Generated by a Set)*
Let M be a nonempty subset of G, and let $T = \{H \leq G \mid M \subseteq H\}$. Then

$$\langle M \rangle = \bigcap_{H \in T} H$$

is the smallest subgroup of G containing M, called the subgroup generated by M.

Definition 1.11. *(Cyclic Group)*
If $\langle M \rangle = \langle a \rangle$ for a single element a, then $\langle a \rangle$ is the cyclic group generated by a.

Example.
$(\mathbb{Z}, +) = \langle 1 \rangle$ and $(\mathbb{Z}_n, +) = \langle \bar{1} \rangle$.
The symmetric group S_3 has $\{(12), (13)\}$ as a generating set.
Every generating set of S_n has at least $n - 1$ elements.

1.2 Coset and Conjugacy

Definition 1.12. *(Equivalence Relation)*
A relation R on a nonempty set S is an equivalence relation if it is reflexive, symmetric, and transitive.

Definition 1.13. *(Coset)*
Let $H \leq G$ and $a \in G$. The sets

$$aH = \{ah \mid h \in H\}, \quad Ha = \{ha \mid h \in H\}$$

are the left coset and the right coset of H represented by a.

Remark.
- A coset is a subgroup only in the case $a \in H$.
- Distinct elements of G may determine the same coset.
- $aH = bH$ if and only if $a^{-1}b \in H$.
- Either $aH = bH$ or $aH \cap bH = \varnothing$.
- $|aH| = |bH|$.
- A left coset need not be a right coset.

Theorem 1.1. *(Lagrange)*
If $H \leq G$ and G is finite, then $|H|\ [G : H] = |G|$, where the index $[G : H]$ is the number of left cosets of H.

Proposition 1.5. *(Product Formula)*
Let $A, B \leq G$, with A and B finite. Then

$$|AB| = \frac{|A|\ |B|}{|A \cap B|}.$$

Definition 1.14. *(Order of an Element)*
The order of $a \in G$ is the smallest positive integer m such that $a^m = e$. If no such m exists, one sets $o(a) = \infty$.

Remark.
Let $a, b \in G$.
- If $o(a) < \infty$, then $o(a) \mid |G|$.
- If $a^t = e$, then $o(a) \mid t$.
- If $o(a) = m < \infty$, $o(b) = n < \infty$, $ab = ba$, and $\gcd(m, n) = 1$, then $o(ab) = mn$.

Remark.
- $\langle a \rangle = \langle a^{-1} \rangle$.
- If G is finite, then $G = \langle a \rangle$ if and only if $|G| = o(a)$.
- A cyclic group of order n has $\varphi(n)$ generators.
- Every subgroup of a cyclic group is cyclic.

Corollary 1.1.
A group of prime order is cyclic, hence abelian.

Definition 1.15. *(Conjugacy)*
Elements $a, b \in G$ are conjugate if $b = g^{-1}ag$ for some $g \in G$. The equivalence class of a is the conjugacy class $[a]$.

Theorem 1.2.

$$|[a]| = [G : C_G(a)].$$

Theorem 1.3. *(Class Equation)*
Let R be a set of representatives of the conjugacy classes of G. Then

$$|G| = |C(G)| + \sum_{a \in R \setminus C(G)} \frac{|G|}{|C_G(a)|}.$$

Definition 1.16. *(p-Group)*
A group of order p^n, with p prime and $n \geq 1$, is a p-group.

Theorem 1.4.
A p-group has nontrivial center.

Proof.
If $|C(G)| = 1$, the class equation gives

$$\sum_{a \in R \setminus C(G)} \frac{|G|}{|C_G(a)|} = p^n - 1.$$

Each index $\frac{|G|}{|C_G(a)|}$ is a positive power of p, so the left side is divisible by p, while $p^n - 1$ is not. ■

Corollary 1.2.
A group of order p^2 is abelian.

Proof.
The center is nontrivial, so $|C(G)|$ is p or p^2. If $|C(G)| = p$ and $a \notin C(G)$, then

$C(G) \subseteq C_G(a)$ and the containment is proper, hence $|C_G(a)| = p^2$. Thus $a \in C(G)$, a contradiction. Therefore $C(G) = G$. ■

Definition 1.17. *(Conjugate Subset)*
For $A \subseteq G$ and $g \in G$, the set $g^{-1}Ag$ is a conjugate of A.

Theorem 1.5.
The number of distinct conjugates of a subset $A \subseteq G$ is $[G : N_G(A)]$.

1.3 Group Homomorphism

Definition 1.18. *(Homomorphism)*
A map $f : G \to H$ is a group homomorphism if $f(ab) = f(a)f(b)$ for all $a, b \in G$. It is a monomorphism if it is injective, an epimorphism if it is surjective, and an isomorphism if it is bijective.

Definition 1.19. *(Isomorphic Groups)*
If there exists an isomorphism $f : G \to H$, one denotes $G \cong H$.

Theorem 1.6.
An infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$, and a cyclic group of order n is isomorphic to $(\mathbb{Z}_n, +)$.

Proof.
Let $G = \langle a \rangle$. The map $\pi : (\mathbb{Z}, +) \to G$ given by $m \mapsto a^m$ is a surjective homomorphism, because $\pi(m + n) = a^{m+n} = \pi(m)\pi(n)$. It is injective when G is infinite. When $|G| = n$, it induces an isomorphism $\overline{m} \mapsto a^m$ from $\mathbb{Z}_n$ onto G. ■

Definition 1.20. *(Normal Subgroup)*
A subgroup $H \leq G$ is normal, denoted by $H \lhd G$, if $g^{-1}hg \in H$ for all $h \in H$ and $g \in G$.

Lemma 1.1.
For $H \leq G$, the following are equivalent.

- $H \lhd G$.
- $g^{-1}Hg \subseteq H$ for every $g \in G$.
- $g^{-1}Hg = H$ for every $g \in G$.
- $gH = Hg$ for every $g \in G$.
- The collection of left cosets equals the collection of right cosets.

Remark.
Normality is not transitive: $H_1 \lhd G$ and $H_2 \lhd H_1$ do not imply $H_2 \lhd G$.

Remark.
If $aH = a'H$ and $bH = b'H$, then $a^{-1}a', b^{-1}b' \in H$, hence

$$(ab)^{-1}(a'b') = \big(b^{-1}(a^{-1}a')b\big)\big(b^{-1}b'\big) \in H.$$

Thus $(ab)H = (a'b')H$. Associativity is inherited from G. Since $Hb = bH$, the same product is the setwise product $(aH)(bH) = a(Hb)H = (ab)H$.

Definition 1.21. *(Quotient Group)*
If $H \lhd G$, the set G/H of left cosets is a group under the operation $(aH)(bH) = (ab)H$. The identity is H, the inverse of aH is $a^{-1}H$, and one denotes $\overline{g} = gH$.

Proposition 1.6.
A subgroup of index 2 is normal.

Definition 1.22. *(Kernel and Image)*
For a homomorphism $f : G \to \tilde{G}$,

$$\ker f = \{x \in G \mid f(x) = e\}, \quad \operatorname{Im} f = \{f(x) \mid x \in G\}.$$

Remark.
$\ker f \lhd G$ and $\operatorname{Im} f \leq \tilde{G}$.

Example.
The homomorphisms $f : \mathbb{Z}_{12} \to \mathbb{Z}_{18}$ are determined by $f\big(\overline{1}\big) = \overline{a}$, where $12\overline{a} = \overline{0}$. Thus $o(\overline{a})$ divides $\gcd(12, 18) = 6$, so the possible images of $\overline{1}$ are

$$\overline{0}, \overline{9}, \overline{6}, \overline{12}, \overline{3}, \overline{15}.$$

Theorem 1.7. *(First Isomorphism Theorem)*
If $f : G \to \tilde{G}$ is a homomorphism, then $\operatorname{Im} f \cong G/\ker f$ by $f(x) \mapsto x \ker f$. Equivalently, if $g : G \to K$ is a homomorphism with $g(\ker f) = \{e\}$, then there is a unique homomorphism $\tilde{g} : G/\ker f \to K$ such that $\tilde{g} \circ \pi = g$, where $\pi : G \to G/\ker f$ is the quotient map.

Definition 1.23. *(Canonical Homomorphism)*
If $N \lhd G$, the canonical homomorphism is $\pi : G \to G/N$, $x \mapsto xN$.

Theorem 1.8. *(Correspondence Theorem)*
Let $N \lhd G$. The map $H \mapsto H/N$ is a bijection between the subgroups of G containing N and the subgroups of G/N, and it preserves normality.

Theorem 1.9. *(Second Isomorphism Theorem)*
Let $N \lhd G$ and $H \leq G$. Then $HN = NH \leq G$, $H \cap N \lhd H$, and

$$HN/N \cong H/(H \cap N).$$

Theorem 1.10. *(Third Isomorphism Theorem)*
Let $A \lhd G$, $B \lhd G$, and $A \leq B$. Then $B/A \lhd G/A$, and

$$(G/A)/(B/A) \cong G/B.$$

Definition 1.24. *(Permutation Group)*
Elements of S_n are permutations, and subgroups of S_n are permutation groups. For an arbitrary set Ω, subgroups of $S(\Omega)$ are transformation groups.

Theorem 1.11. *(Cayley)*
Every group is isomorphic to a transformation group. Every finite group is isomorphic to a permutation group.

1.4 Permutation and Group Action

Definition 1.25. *(Even and Odd Permutations)*
A permutation is even if it is a product of an even number of transpositions, and odd otherwise.

Definition 1.26. *(Alternating Group)*
A_n is the set of all even permutations in S_n. Then $A_n \leq S_n$ and $|A_n| = \frac{|S_n|}{2}$.

Remark.
The index $[S_n : A_n] = 2$, so $A_n \lhd S_n$.

Definition 1.27. *(Group Action)*
A group G acts on a set Ω if there is a homomorphism $\rho : G \to S(\Omega)$. The kernel of the action is

$$\ker \rho = \{g \in G \mid \rho(g) = \mathrm{id}_\Omega\}.$$

The action is faithful if $\ker \rho = \{e\}$.

Definition 1.28. *(G-Set)*
A set Ω is a G-set if there is a map $G \times \Omega \to \Omega$, $(g, x) \mapsto gx$, such that $ex = x$ and $(gh)x = g(hx)$ for all $x \in \Omega$ and $g, h \in G$.

Definition 1.29. *(Orbit and Stabilizer)*
The orbit and the stabilizer of $x \in \Omega$ are

$$O_x = \{gx \mid g \in G\}, \quad G_x = \{g \in G \mid gx = x\}.$$

The stabilizer is a subgroup of G. The action is transitive if Ω has only one orbit.

Remark.
For $x, y \in \Omega$, either $O_x = O_y$ or $O_x \cap O_y = \varnothing$. Thus Ω is the disjoint union of the orbits.

Lemma 1.2.
The map $O_x \to G/G_x$, $gx \mapsto gG_x$, is a bijection.

Theorem 1.12. *(Orbit-Stabilizer)*
If G acts on Ω, then

$$|O_x| = [G : G_x] = \frac{|G|}{|G_x|}$$

whenever the orbit is finite. If the action is transitive, then $|\Omega| = \frac{|G|}{|G_x|}$.

Theorem 1.13. *(Burnside)*
Let a finite group G act on a finite set Ω, and let t be the number of orbits. If $F_g = |\{x \in \Omega \mid gx = x\}|$, then

$$t = \frac{1}{|G|} \sum_{g \in G} F_g.$$

Example.
How many necklaces can be made from six beads, two of each of three colors, up to rotation and reflection?

Solution.
The symmetry group is the dihedral group D_6 of order 12. The identity fixes $\frac{6!}{(2!)^3} = 90$ colorings. The rotations by 1, 2, 4, and 5 positions fix none, the rotation by 3 positions fixes 6, and each of the six reflections fixes 6. Hence

$$t = \frac{1}{12}(90 + 6 + 6 \cdot 6) = 11.$$

1.5 Sylow Theorem

Definition 1.30. *(Simple Group)*
A group $G \neq \{e\}$ is simple if its only normal subgroups are $\{e\}$ and G.

Theorem 1.14. *(Sylow I)*
Let G be a finite group, p a prime, and $r \geq 1$. If p^r divides $|G|$, and $N(p^r)$ denotes the number of subgroups of order p^r, then

$$N(p^r) \equiv 1 (\mathrm{mod}\, p).$$

In particular, G has a subgroup of order p^r.

Definition 1.31. *(Sylow p-Subgroup)*
Let $|G| = p^m n$ with p prime, $m \geq 1$, and p not dividing n. A subgroup of order p^m is a Sylow p-subgroup of G.

Theorem 1.15. *(Sylow II)*
Any two Sylow p-subgroups of a finite group are conjugate.

Remark.
If P is a Sylow p-subgroup, then

$$N(p^m) = kp + 1 = \frac{|G|}{|N_G(P)|}.$$

In particular, $k = 0$ means $P \lhd G$, since every conjugate of P is again a Sylow p-subgroup.

Theorem 1.16. *(Sylow III)*
Every p-subgroup of a finite group is contained in some Sylow p-subgroup.

Example.
A group G of order 35 is cyclic.

Proof.
Indeed $N(5) = 5k + 1$ divides 7, and $N(7) = 7k' + 1$ divides 5, so there is a unique Sylow 5-subgroup $H = \langle a \rangle$ and a unique Sylow 7-subgroup $K = \langle b \rangle$. Both are normal, and $H \cap K = \{e\}$, so

$$a^{-1}b^{-1}ab \in H \cap K.$$

Thus $ab = ba$. The element ab has order 35, so $G = \langle ab \rangle$. ■

Example.
No group of order $148 = 2^2 \cdot 37$ is simple.

Proof.
Here $N(37) = 37k + 1$ divides 4, so $N(37) = 1$ and the Sylow 37-subgroup is normal. ■

Example.
No group of order $56 = 2^3 \cdot 7$ is simple.

Proof.
The possibilities are $N(7) \mid 8$ with $N(7) \equiv 1 (\mathrm{mod}\, 7)$, and $N(8) \mid 7$ with $N(8) \equiv 1 (\mathrm{mod}\, 2)$, so $N(7) \in \{1, 8\}$ and $N(8) \in \{1, 7\}$. These two larger values cannot occur

together. If $N(7) = 1$, the Sylow 7-subgroup is normal. If $N(7) = 8$, the eight Sylow 7-subgroups meet pairwise at $\{e\}$ and cover $6 \cdot 8 + 1 = 49$ elements. The remaining $56 - 49 = 7$ elements, together with e, form a unique Sylow 2-subgroup, which is normal. ■

Example.
A group of order 6 is isomorphic to $\mathbb{Z}_6$ or to S_3.

Proof.
Here $N(3) = 1$, while $N(2)$ is 1 or 3. If $N(2) = 1$, take $o(a) = 3$ and $o(b) = 2$. Both Sylow subgroups are normal and intersect trivially, so $ab = ba$ and $o(ab) = 6$. Thus $G \cong \mathbb{Z}_6$. If $N(2) = 3$, conjugation on the three Sylow 2-subgroups gives a homomorphism $f : G \to S_3$. The action is transitive, so $\ker f \neq G$. A kernel of order 2 would be a normal Sylow 2-subgroup, and a kernel of order 3 would normalize every Sylow 2-subgroup, forcing one of them to be normal. Either case contradicts $N(2) = 3$. Hence f is an isomorphism and $G \cong S_3$. ■

Example.
For a prime p and $m \geq 2$, no group of order p^m is simple.

Proof.
The class equation gives $|C(G)| \neq 1$, and $C(G) \lhd G$. If the center is proper it is a nontrivial proper normal subgroup. If $C(G) = G$, then G is abelian and any subgroup of order p is a nontrivial proper normal subgroup. ■

Example.
If p and q are distinct primes, no group of order pq is simple.

Proof.
Assume $p < q$. Then $N(q) \equiv 1 (\mathrm{mod}\, q)$ and $N(q)$ divides p, so $N(q) = 1$. ■

Example.
If p and q are distinct primes, no group of order $p^2 q$ is simple.

Proof.
One has $N(p^2) = kp + 1$ dividing q, and $N(q) = k'q + 1$ dividing p^2. If $p > q$, then $N(p^2) = 1$. If $p < q$, then $N(q)$ cannot be p, because $p \equiv 1 (\mathrm{mod}\, q)$ is impossible, so $N(q)$ is 1 or p^2. In the latter case the Sylow q-subgroups contribute $p^2(q-1) + 1$ elements, and the remaining $p^2 - 1$ non-identity elements form a unique subgroup of order p^2. Thus $N(p^2) = 1$. ■

Example.
No group of order 24 is simple.

Proof.
Let G act by conjugation on its Sylow 3-subgroups. If there is only one, it is normal. If there are four, the action gives a homomorphism $f : G \to S_4$. The kernel is a proper normal subgroup. It is nontrivial, or else f is injective and $G \cong S_4$, which is not simple. ■

Theorem 1.17. *(Frattini)*
Let $N \lhd G$, and let P be a Sylow p-subgroup of N. Then $G = NN_G(P)$.

Proof.
For $g \in G$ one has $gPg^{-1} \leq N$. This conjugate is a Sylow p-subgroup of N, so $nPn^{-1} = gPg^{-1}$ for some $n \in N$. Then $g^{-1}n \in N_G(P)$, and hence $g \in NN_G(P)$. ■

1.6 Finitely Generated Abelian Group

Definition 1.32. *(External Direct Product)*
Let G_1 and G_2 be groups. The set $G_1 \times G_2$ becomes a group under

$$(a_1, b_1)(a_2, b_2) = (a_1a_2, b_1b_2).$$

It is the external direct product, denoted by $G_1 \times G_2$. If the groups are additive, one denotes $G_1 \oplus G_2$.

Remark.
Set $\tilde{G}_1 = G_1 \times \{e_2\}$ and $\tilde{G}_2 = \{e_1\} \times G_2$. Then $\tilde{G}_i \cong G_i$, $\tilde{G}_i \lhd G_1 \times G_2$, and $\tilde{G}_1\tilde{G}_2 = G_1 \times G_2$.

Definition 1.33. *(Internal Direct Product)*
Let $H, K \lhd G$. If $G = HK$ and $H \cap K = \{e\}$, then G is the internal direct product of H and K.

Definition 1.34. *(Quaternion Group)*
The quaternion group is

$$Q_8 = \{\pm e, \pm i, \pm j, \pm k\},$$

with $i^2 = j^2 = k^2 = -e$, $ij = k = -ji$, $jk = i = -kj$, and $ki = j = -ik$.

Remark.
As matrices,

$$Q_8 \cong \left\{ \pm \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \pm \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \pm \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \pm \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \right\}.$$

Definition 1.35. *(Free Abelian Group)*
The group $\mathbb{Z}^n$ is the free abelian group of rank n. If a finitely generated free abelian group G has a basis of r elements, then r is the rank of G.

Example.

$$\langle a, b \mid a^n = 1, b^2 = 1, ba = a^{-1}b \rangle = D_n,$$
$$\langle a, b \mid a^4 = 1, a^2 = b^2, ba = a^3 b \rangle = Q_8.$$

Here D_n is the dihedral group of order $2n$.

Lemma 1.3.
Let G be a finite abelian group of order $n = p_1^{m_1} \cdots p_t^{m_t}$. Then

$$G = G_1 \oplus \cdots \oplus G_t,$$

where G_i is the Sylow p_i-subgroup of G.

Theorem 1.18. *(Fundamental Theorem of Finite Abelian Groups)*
Let G be an abelian group of order $n = p_1^{m_1} \cdots p_t^{m_t}$. Write each $m_i = m_{i1} + \cdots + m_{ik_i}$ with positive integers m_{ij}. Then

$$G \cong \oplus_{i=1}^{t} \left(\mathbb{Z}_{p_i^{m_{i1}}} \oplus \cdots \oplus \mathbb{Z}_{p_i^{m_{ik_i}}} \right).$$

The prime-power orders $p_i^{m_{ij}}$ are the elementary divisors of G. Equivalently, there exist positive integers $d_1 \mid d_2 \mid \cdots \mid d_s$ with $d_1 \cdots d_s = n$ such that

$$G \cong \mathbb{Z}_{d_1} \oplus \cdots \oplus \mathbb{Z}_{d_s}.$$

The integers $d_1, \ldots, d_s$ are the invariant factors of G.

Example.
How many abelian groups of order 900 are there up to isomorphism?

Solution.
Write $900 = 2^2 \cdot 3^2 \cdot 5^2$. An abelian group of order p^2 is isomorphic to $\mathbb{Z}_{p^2}$ or to $\mathbb{Z}_p \oplus \mathbb{Z}_p$, so there are $2^3 = 8$ isomorphism types. By elementary divisors, and the corresponding invariant factors, they are

$$\mathbb{Z}_4 \oplus \mathbb{Z}_9 \oplus \mathbb{Z}_{25} \cong \mathbb{Z}_{900},$$
$$\mathbb{Z}_4 \oplus \mathbb{Z}_9 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_5 \cong \mathbb{Z}_5 \oplus \mathbb{Z}_{180},$$
$$\mathbb{Z}_4 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_{25} \cong \mathbb{Z}_3 \oplus \mathbb{Z}_{300},$$
$$\mathbb{Z}_4 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_5 \cong \mathbb{Z}_{15} \oplus \mathbb{Z}_{60},$$
$$\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_9 \oplus \mathbb{Z}_{25} \cong \mathbb{Z}_2 \oplus \mathbb{Z}_{450},$$
$$\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_9 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_5 \cong \mathbb{Z}_{10} \oplus \mathbb{Z}_{90},$$
$$\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_{25} \cong \mathbb{Z}_6 \oplus \mathbb{Z}_{150},$$
$$\mathbb{Z}_2 \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus \mathbb{Z}_5 \oplus \mathbb{Z}_5 \cong \mathbb{Z}_{30} \oplus \mathbb{Z}_{30}.$$

2 Ring

2.1 Ring Basics

Definition 2.1. *(Ring)*
A ring is a set R with two binary operations $+$ and $\cdot$ such that
- $(R, +)$ is an abelian group, whose identity 0 is the zero element;
- $(R, \cdot)$ is a semigroup;
- multiplication distributes over addition on both sides, namely $a(b + c) = ab + ac$ and $(b + c)a = ba + ca$.

If multiplication is commutative, R is a commutative ring. The additive inverse of a is written $-a$.

Example.
The polynomial ring is $R[x] = \{a_n x^n + \cdots + a_1 x + a_0 \mid a_i \in R\}$.

Proposition 2.1.
In a ring, $0a = 0 = a0$, $(-a)b = -ab = a(-b)$, and $(na)b = n(ab) = a(nb)$. An identity element, if it exists, is unique.

Definition 2.2. *(Zero Divisor)*
A nonzero element a is a left zero divisor if $ab = 0$ for some $b \neq 0$, and a right zero divisor if $ba = 0$ for some $b \neq 0$. Both are called zero divisors.

Proposition 2.2.
Cancellation holds in a ring without zero divisors: $ab = ac, a \neq 0$ implies $b = c$.

Definition 2.3. *(Integral Domain)*
An integral domain is a commutative ring with identity and with no zero divisors.

Definition 2.4. *(Unit)*
Let R be a nonzero ring with identity. An element with a multiplicative inverse is a unit of R. The set of all units is $U(R)$.

Definition 2.5. *(Division Ring and Field)*
A nonzero ring R with identity is a division ring if $U(R) = R \setminus \{0\}$.
A commutative division ring is a field.

Definition 2.6. *(Subring)*
A nonempty subset S of a ring R is a subring if S with the operations of R is a ring.

Proposition 2.3.
A nonempty subset S is a subring if and only if $a - b \in S$ and $ab \in S$ for all $a, b \in S$.

Proposition 2.4.
A subset F of a field K is a subfield if and only if $a - b \in F$ and $ab^{-1} \in F$ whenever $a, b \in F$ and $b \neq 0$.

Definition 2.7. *(Ring Homomorphism)*
A map $f : R \to R'$ is a ring homomorphism if $f(a + b) = f(a) + f(b)$ and $f(ab) = f(a)f(b)$ for all $a, b \in R$.

Remark.
A ring homomorphism need not send the identity to the identity.

2.2 Ideal and Quotient Ring

Definition 2.8. *(Ideal)*
Let I be a subgroup of $(R,+)$. If $ra \in I$ for all $a \in I$ and $r \in R$, then I is a left ideal. If $ar \in I$, then I is a right ideal. An ideal, denoted by $I \lhd R$, means a two-sided ideal.

Remark.
An ideal is a subring. If I_1 and I_2 are ideals, then so are $I_1 \cap I_2$, $I_1 + I_2$, and

$$I_1 I_2 = \left\{ \sum_{i=1}^{n} r_i s_i \mid r_i \in I_1, s_i \in I_2 \right\}.$$

Definition 2.9. *(Principal Ideal and Principal Ideal Domain)*
An ideal generated by one element is a principal ideal. An integral domain in which every ideal is principal is a principal ideal domain (PID).

Definition 2.10. *(Quotient Ring)*
If $I \lhd R$, the quotient group R/I is a ring under the multiplication $(a+I)(b+I) = ab + I$.

Example.

$$\mathbb{Q}[x]/\langle x^2 - 2\rangle \cong \mathbb{Q}\left[\sqrt{2}\right] = \left\{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\right\}.$$

The map $a + bx + \langle x^2 - 2\rangle \mapsto a + b\sqrt{2}$ is a well-defined bijection, and it is the isomorphism induced by the evaluation homomorphism $\mathbb{Q}[x] \to \mathbb{Q}\left[\sqrt{2}\right]$, whose kernel is $\langle x^2 - 2\rangle$.

Theorem 2.1. *(First Isomorphism Theorem)*
If $f : R \to R'$ is a ring homomorphism, then $\ker f \lhd R$ and $\operatorname{Im} f \cong R/\ker f$.

Theorem 2.2. *(Second and Third Isomorphism Theorems)*
Let $I \lhd R$ and let S be a subring of R. Then $I \cap S \lhd S$, $I \lhd I + S$, and

$$(I+S)/I \cong S/(I \cap S).$$

If also $J \lhd R$ and $I \subseteq J$, then $J/I \lhd R/I$ and

$$(R/I)/(J/I) \cong R/J.$$

Definition 2.11. *(Simple Ring)*
A ring whose only ideals are $\{0\}$ and itself is a simple ring.

Definition 2.12. *(Prime Ideal)*
A proper ideal P of R is prime if, whenever I and J are ideals with $IJ \subseteq P$, one has $I \subseteq P$ or $J \subseteq P$.

Remark.
Let R be a commutative ring with identity, and let P be a proper ideal. The following are equivalent.

- P is prime.
- Whenever $ab \in P$, one has $a \in P$ or $b \in P$.
- R/P is an integral domain.

Definition 2.13. *(Maximal Ideal)*
A proper ideal M is maximal if $M \subseteq I \lhd R$ implies $I = M$ or $I = R$.

Proposition 2.5.
Let R be a commutative ring with identity, and let P be an ideal. Then P is maximal if and only if R/P is a field. In particular, a maximal ideal is prime.

Proposition 2.6.
In a nonzero ring with identity, every proper ideal is contained in a maximal ideal.

Definition 2.14. *(Characteristic)*
If there is a positive integer m such that $mr = 0$ for every $r \in R$, the smallest such m is the characteristic of R, denoted by $\operatorname{char} R$. Otherwise $\operatorname{char} R = 0$.

Proposition 2.7.
In a ring with identity and without zero divisors, every nonzero element has the same additive order. That order is either infinite or prime, and it equals $\operatorname{char} R = o(1)$.

Proof.
Let e be the identity and $n = o(e)$. For $r \neq 0$, if $mr = 0$, then $(me)r = 0$, so $me = 0$ and $n \mid m$. Also $nr = (ne)r = 0$, so the additive order of r divides n. Thus every nonzero element has additive order n. If $n = st$, then $(se)(te) = ne = 0$, so $se = 0$ or $te = 0$. Hence n is prime or infinite. ■

Proposition 2.8.
If R has an identity e, then $\varphi : \mathbb{Z} \to R$, $n \mapsto ne$, is a ring homomorphism.

Remark.
If $\operatorname{char} R = 0$, then R contains a subring isomorphic to $\mathbb{Z}$.

Proposition 2.9. *(Frobenius)*
Let R be a commutative ring of prime characteristic p. Then $f : R \to R$, $r \mapsto r^p$, is a ring homomorphism.

Definition 2.15. *(Field of Fractions)*
Let D be an integral domain and $D^* = D \setminus \{0\}$. On $D \times D^*$, declare $(r, s) \sim (r', s')$ when $rs' = r's$. The equivalence classes, written r/s, form a field K under the usual addition and multiplication of fractions. This field is the field of fractions of D.

Remark.
The field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$. The field of fractions of $R[x]$ is

$$R(x) = \{f/g \mid f, g \in R[x], g \neq 0\}.$$

In particular $\mathbb{Q}\big(\sqrt{2}\big) = \mathbb{Q}\big[\sqrt{2}\big]$.

Proposition 2.10.
A field of characteristic 0 contains a subfield isomorphic to $\mathbb{Q}$. A field of characteristic $p > 0$ contains a subfield isomorphic to $\mathbb{Z}_p$.

Definition 2.16. *(Direct Sum of Rings)*
On $R_1 \times \cdots \times R_n$ one defines

$$(a_1, ..., a_n) + (b_1, ..., b_n) = (a_1 + b_1, ..., a_n + b_n),$$

and likewise for multiplication, coordinate by coordinate. The resulting ring is the direct sum $R_1 \oplus \cdots \oplus R_n$.

Proposition 2.11.
$R \cong R_1 \oplus \cdots \oplus R_n$ if and only if R has ideals $I_i \cong R_i$ such that every element of R can be written uniquely as $r = r_1 + \cdots + r_n$ with $r_i \in I_i$. One then says that R is the internal direct sum of $I_1, ..., I_n$.

Definition 2.17. *(Comaximal Ideals)*
Ideals I and J are comaximal if $I + J = R$.

Theorem 2.3. *(Chinese Remainder Theorem)*
Let $I_1, ..., I_n$ be pairwise comaximal ideals of a ring R with identity. Then

$$R/(I_1 \cap \cdots \cap I_n) \cong R/I_1 \oplus \cdots \oplus R/I_n,$$

by $r + (I_1 \cap \cdots \cap I_n) \mapsto (r + I_1, ..., r + I_n)$.

Example.
The maximal ideals of $\mathbb{Z}_8 \oplus \mathbb{Z}_{15}$ are $2\mathbb{Z}_8 \oplus \mathbb{Z}_{15}$, $\mathbb{Z}_8 \oplus 3\mathbb{Z}_{15}$, and $\mathbb{Z}_8 \oplus 5\mathbb{Z}_{15}$.

2.3 Unique Factorization

Definition 2.18. *(Divisibility)*
Let D be an integral domain and $a, b \in D$. If $a = bc$ for some $c \in D$, then b divides a, written $b \mid a$. If $b \mid a$ and $a \mid b$, then a and b are associates, written $a \sim b$.

Remark.
$b \mid a$ if and only if $\langle a \rangle \subseteq \langle b \rangle$, and $a \sim b$ if and only if $\langle a \rangle = \langle b \rangle$.

Definition 2.19. *(Irreducible Element)*
A nonzero nonunit $p \in D$ is irreducible if $p = ab$ implies that a is a unit or b is a unit. Equivalently, $b \mid p$ implies $b \in U(D)$ or $b \sim p$.

Definition 2.20. *(Prime Element)*
A nonzero nonunit $p \in D$ is prime if $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Lemma 2.1.
In an integral domain, every prime element is irreducible.

Definition 2.21. *(Norm)*
For $a + b\sqrt{d} \in \mathbb{Q}\left[\sqrt{d}\right]$, the norm is $N\left(a + b\sqrt{d}\right) = |a^2 - db^2|$.

Lemma 2.2.
Let D be an integral domain and $p \in D \setminus \{0\}$. Then p is prime if and only if $\langle p \rangle$ is a prime ideal.

Definition 2.22. *(Unique Factorization Domain)*
An integral domain D is a unique factorization domain (UFD) if every nonzero nonunit $a \in D$ can be written as a product of irreducible elements, and the factorization is unique in the following sense. If $a = p_1 \cdots p_n = q_1 \cdots q_m$ with each factor irreducible, then $n = m$ and, after reordering, $p_i \sim q_i$ for every i.

Theorem 2.4.
In a UFD, every irreducible element is prime.

Definition 2.23. *(Factor Chain)*
A sequence $a_1, a_2, \ldots$ in an integral domain is a factor chain if each a_i is a proper divisor of a_{i-1}, for $i \geq 2$.

Theorem 2.5.
Every factor chain in a UFD is finite.

Lemma 2.3.
An integral domain is a UFD if every nonzero nonunit has a finite factor chain and every irreducible element is prime.

Definition 2.24. *(Standard Factorization)*
In a UFD, every nonzero a may be written $a = \eta p_1^{r_1} \cdots p_s^{r_s}$, where η is a unit, the p_i are pairwise nonassociate irreducible elements, and each r_i is a positive integer.

Definition 2.25. *(Greatest Common Divisor)*
Let $a, b \in D$ be nonzero. An element d is a greatest common divisor of a and b if $d \mid a$, $d \mid b$, and every common divisor of a and b divides d.

Lemma 2.4.
If every pair of elements in an integral domain has a greatest common divisor, then every irreducible element is prime.

Theorem 2.6.
In a UFD, every pair of elements has a greatest common divisor.

Proposition 2.12.
Every factor chain in a PID is finite.

Proposition 2.13.
Let D be a PID and let $a \in D \setminus \{0\}$. The following are equivalent.
- a is prime.
- a is irreducible.
- $\langle a \rangle$ is a maximal ideal.
- $\langle a \rangle$ is a prime ideal.

Theorem 2.7.
Every PID is a UFD.

Theorem 2.8. *(Gauss)*
If D is a UFD, then so is $D[x]$.

Definition 2.26. *(Indeterminate)*
Let R be a ring with identity, and let $\tilde{R}$ be a ring containing R as a subring. An element $x \in \tilde{R}$ is an indeterminate over R if $rx = xr$ for every $r \in R$, and whenever $a_0, \ldots, a_n \in R$ are not all zero,

$$a_0 + a_1 x + \cdots + a_n x^n \neq 0.$$

Proposition 2.14.
Every commutative ring with identity admits an indeterminate. The resulting polynomial ring is $R[x]$.

Corollary 2.1. *(Remainder Theorem)*
Let R be a commutative ring with identity, $f \in R[x]$, and $c \in R$. There is a unique $q \in R[x]$ such that $f(x) = q(x)(x - c) + f(c)$.

Corollary 2.2.
Over an integral domain, a polynomial of degree n has at most n roots.

Theorem 2.9. *(Eisenstein)*
Let D be a UFD and let $f(x) = a_n x^n + \cdots + a_1 x + a_0 \in D[x]$ have content 1, that is $\gcd(a_0, \ldots, a_n) = 1$. If a prime element $p \in D$ divides a_i for every $i < n$, does not divide a_n, and p^2 does not divide a_0, then f is irreducible in $D[x]$.

Remark.
$\mathbb{Z}_2[x]/\langle x^3 + x + 1 \rangle$ and $\mathbb{Z}_2[x]/\langle x^3 + x^2 + 1 \rangle$ are fields with eight elements.

Definition 2.27. *(Euclidean Domain)*
An integral domain D is a Euclidean domain if there is a function $\varphi : D \to \mathbb{N}$ such that $\varphi(x) = 0$ if and only if $x = 0$, and for all $a, b \in D$ with $b \neq 0$ there exist $q, r \in D$ with

$$a = qb + r, \quad \varphi(r) < \varphi(b).$$

Proposition 2.15.
Every Euclidean domain is a PID, and hence a UFD.

Remark.
The classes of rings satisfy

$$\text{field} \subsetneq \text{Euclidean domain} \subsetneq \text{PID} \subsetneq \text{UFD} \subsetneq \text{integral domain}.$$

3 Field

3.1 Field Extension

Definition 3.1. *(Field Extension)*
If F is a subfield of a field K, then K is an extension of F, denoted by K/F.

Definition 3.2. *(Algebraic and Transcendental Elements)*
An element u of some extension of F is algebraic over F if it is a root of a nonzero polynomial in $F[x]$, and transcendental over F otherwise. The extension K/F is algebraic if every element of K is algebraic over F, and transcendental otherwise.

Definition 3.3. *(Minimal Polynomial)*
Let u be algebraic over F. The monic polynomial in $F[x]$ of least degree having u as a root is the minimal polynomial of u over F.

Proposition 3.1.
Let α be algebraic over F, and let $u(x) \in F[x]$ be monic. The following are equivalent.

- u is the minimal polynomial of α.
- u is irreducible over F and $u(\alpha) = 0$.
- Whenever $f(\alpha) = 0$, one has $u \mid f$.

Proposition 3.2.
Let K/F and $u \in K$, and write $F[u] = \{f(u) \mid f \in F[x]\}$. If u is algebraic over F with minimal polynomial f, then

$$F[u] = F(u) \cong F[x]/\langle f \rangle.$$

If u is transcendental over F, then $F[u] \cong F[x]$.

Theorem 3.1. *(Tower Law)*
If $F \subseteq K \subseteq E$ are fields and the degrees are finite, then $[E : F] = [E : K][K : F]$.

Corollary 3.1.
Let u and v lie in some extension of F.

- If both are transcendental over F, then there is an isomorphism $\sigma : F(u) \to F(v)$ with $\sigma(u) = v$ and $\sigma \mid_F = \mathrm{id}$.
- If both are algebraic over F, then such an isomorphism exists if and only if u and v have the same minimal polynomial over F.

Theorem 3.2.
Let $f \in F[x]$ have degree $n \geq 1$. There is a simple extension $K = F(u)$ such that
- $u \in K$ is a root of f;
- $[F(u) : F] \leq n$, and equality holds if and only if f is irreducible in $F[x]$;
- if f is irreducible, then K is unique up to an isomorphism that fixes F and sends the adjoined root to any prescribed root.

Definition 3.4. *(Algebraically Closed Field)*
A field K is algebraically closed if every element that is algebraic over K, in any extension of K, already lies in K.

3.2 Splitting Field

Definition 3.5. *(Galois Group)*
For an extension K/F,

$$\mathrm{Gal}(K/F) = \{\sigma \in \mathrm{Aut}(K) \mid \sigma \mid_F = \mathrm{id}\}.$$

Lemma 3.1.
Let u be algebraic over F, with minimal polynomial f. The order $|\mathrm{Gal}(F(u)/F)|$ equals the number of distinct roots of f that lie in $F(u)$.

Definition 3.6. *(Splitting Field)*
Let $f \in F[x]$ have positive degree. An extension E/F is a splitting field of f over F if

$$f(x) = c(x - a_1)\cdots(x - a_n)$$

with $c \in F$ and $a_i \in E$, and $E = F(a_1, \ldots, a_n)$. Thus E is the smallest extension of F containing every root of f.

Lemma 3.2.
If $f \in F[x]$ is irreducible, there is a simple extension $F(u)$, unique up to an isomorphism fixing F, such that u is a root of f.

Proposition 3.3.
Every polynomial of positive degree over F has a splitting field over F.

Definition 3.7. *(Separable Polynomial)*
A polynomial $f \in F[x]$ of positive degree is separable if every irreducible factor of f has no repeated root in a splitting field over F.

Definition 3.8. *(Formal Derivative)*
The formal derivative of $f(x) = a_0 + a_1 x + \cdots + a_n x^n$ is

$$f'(x) = a_1 + 2a_2 x + \cdots + n a_n x^{n-1}.$$

Lemma 3.3.
A polynomial $f \in F[x]$ of positive degree has a repeated root if and only if $\gcd(f, f') \neq 1$. This is a property of f itself and does not depend on the extension of F in which the roots are taken.

Theorem 3.3.
Let $\sigma : F \to F'$ be an isomorphism of fields, let $f \in F[x]$ have positive degree, and let E and E' be splitting fields of f over F and of $\sigma(f)$ over F' respectively. Then σ extends to an isomorphism $E \to E'$. The number m of such extensions satisfies $1 \leq m \leq [E : F]$, and $m = [E : F]$ if and only if f is separable.

Corollary 3.2.
If E is a splitting field of f over F, then $|\operatorname{Gal}(E/F)| \leq [E : F]$.

Example.
The splitting field of $x^5 - 1$ over $\mathbb{Q}$ is the fifth cyclotomic field $E = \mathbb{Q}(\zeta)$, where $\zeta = e^{2\pi i/5}$.

Remark.
Each $\sigma_k \in \mathrm{Gal}(E/\mathbb{Q})$ is determined by $\sigma_k(\zeta) = \zeta^k$ with $\gcd(k, 5) = 1$. The map $\sigma_k \mapsto k$ is an isomorphism

$$\mathrm{Gal}(E/\mathbb{Q}) \cong (\mathbb{Z}_5)^\times \cong \mathbb{Z}_4.$$

3.3 Finite Field

Definition 3.9. *(Finite Field)*
A field with finitely many elements is a finite field, also called a Galois field.

Theorem 3.4.
Let F be a finite field. Then $\mathrm{char}\, F = p$ for some prime p, $|F| = p^n$ where $n = \big[F : \mathbb{Z}_p\big]$, and $(F, +) \cong \mathbb{Z}_p^n$.

Theorem 3.5.
The multiplicative group of a finite field F with p^n elements is cyclic of order $p^n - 1$. Thus $F = \mathbb{Z}_p(u)$ for some algebraic element u of degree n over $\mathbb{Z}_p$.

Definition 3.10. *(Primitive Element)*
A generator of the simple extension $F/\mathbb{Z}_p$ may be taken to be a generator of $F^\times$. Such an element is a primitive element.

Theorem 3.6.
For every prime p and every integer $n \geq 1$, there exists a field of order p^n, and it is unique up to isomorphism.

Definition 3.11. *(Frobenius Automorphism)*
On a field F of order p^n, the map $\sigma_p : F \to F$, $a \mapsto a^p$, is an automorphism of F, called the Frobenius automorphism.

Remark.
The subfields of a field of order p^n are exactly the fields of order p^m with $m \mid n$.